

## Comparison of Optimization Methods on Rosenbrock Function

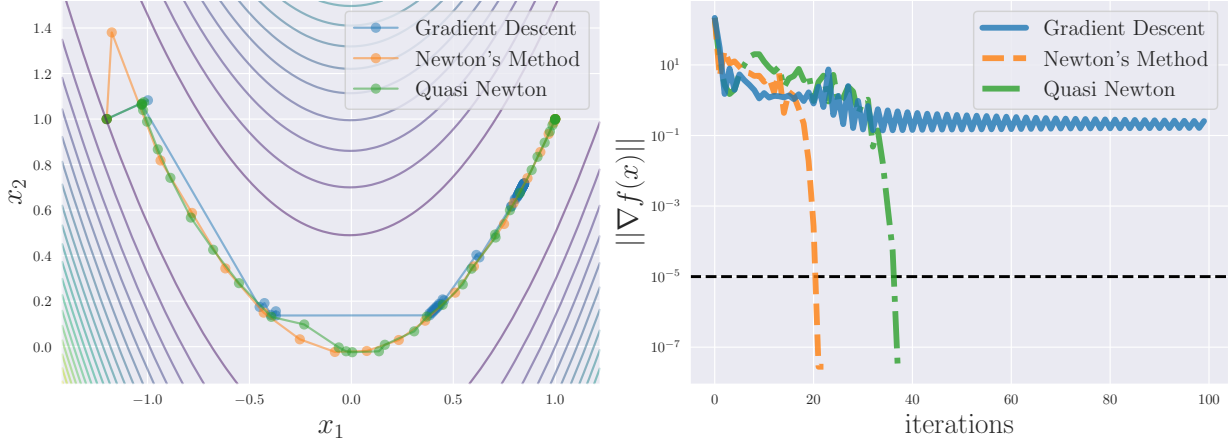


Figure 1: Comparison of Newton's method, quasi-Newton methods, and gradient descent.

## 1 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method but aim to reduce its primary drawback, namely the high computational cost of evaluating the Hessian matrix. Instead of using the true Hessian matrix  $\nabla^2 f(x_k)$ , an approximation matrix  $B_k$  is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 1 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent updates the iterate in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method requires the fewest iterations, but it has the highest computational cost per step. Quasi-Newton methods lie between these two extremes and strike a balance between them. In particular, when the methods are compared in terms of total computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Quasi-Newton methods with line search generate a sequence  $\{x_k\}_k$  defined by

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k).$$

Here,  $\alpha_k > 0$  is the step size determined by line search,  $B_k$  is an approximation of the Hessian matrix  $\nabla^2 f(x_k)$  at the point  $x_k$ , and  $H_k := B_k^{-1}$  denotes its inverse. This update rule closely resembles that of Newton's method, which is why it is called a quasi-Newton method.

The core of quasi-Newton methods lies in how to update  $B_k$  at each iteration so that it approaches the true Hessian matrix. Figure 2 illustrates this concept. First, around the current point  $x_k$ , a quadratic approximation model of the objective function  $f$  is constructed using  $B_k$ . Next, this quadratic model is minimized to obtain the next point  $x_{k+1}$ . After obtaining  $x_{k+1}$ , the approximation matrix  $B_k$  is updated to  $B_{k+1}$  using the gradient information at  $x_k$  and  $x_{k+1}$ . This procedure is repeated until convergence, which constitutes the quasi-Newton methods.

### 1.1 Secant Condition

Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be of class  $C^2$ . Suppose that a symmetric matrix  $B_k$  is given as an approximation of the Hessian matrix  $\nabla^2 f(x_k)$ . We consider updating the approximate Hessian to  $B_{k+1}$  for the next point  $x_{k+1}$ . Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry

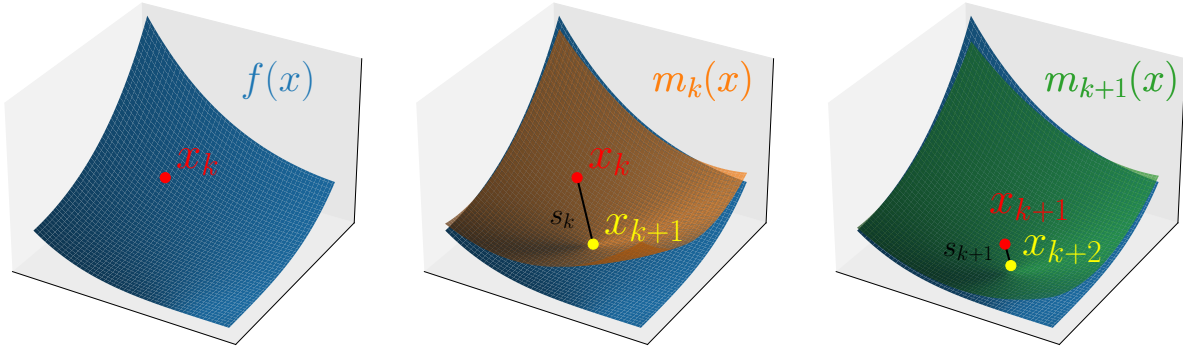


Figure 2: Conceptual illustration of quasi-Newton methods. (1) Given the objective function  $f$  (blue surface) and the current point  $x_k$  (red point). (2) Move to the minimizer  $x_{k+1}$  (yellow cross) of the quadratic model  $m_k(x)$  (orange surface) constructed using the current approximate Hessian. (3) Update the quadratic model (green surface) and repeat this procedure.

on  $B_{k+1}$ , since the true Hessian matrix is symmetric. For the current point  $x_k$  and the next point  $x_{k+1}$ , we define the step and the gradient difference as follows:

$$s_k := x_{k+1} - x_k, \quad y_k := \nabla f(x_{k+1}) - \nabla f(x_k).$$

In the following, we generally assume that  $s_k \neq 0$  and  $y_k^\top s_k \neq 0$ . Note that the  $s$  in  $s_k$  stands for “step.” Since the new approximation is constructed around  $x_{k+1}$ , we approximate  $\nabla f(x_k)$  using the Hessian matrix  $\nabla^2 f(x_{k+1})$  and its approximation  $B_{k+1}$  as follows:

$$\begin{aligned} \nabla f(x_k) &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}). \end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k).$$

Equivalently, this can be rewritten as

$$B_{k+1}s_k = y_k.$$

This relation is called the secant condition or the quasi-Newton equation [4].

More precisely, the secant condition can also be justified as follows. Consider a quadratic approximation model around  $x_{k+1}$ :

$$m_{k+1}(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}(x - x_{k+1}).$$

By construction, regardless of  $B_{k+1}$ , this model satisfies

$$\begin{cases} m_{k+1}(x_{k+1}) = f(x_{k+1}), \\ \nabla m_{k+1}(x_{k+1}) = \nabla f(x_{k+1}). \end{cases}$$

In addition, the secant condition ensures that the model satisfies

$$\nabla m_{k+1}(x_k) = \nabla f(x_{k+1}) - B_{k+1}(x_{k+1} - x_k) \quad (\text{definition of the model})$$

$$\begin{aligned}
&= \nabla f(x_{k+1}) - (\nabla f(x_{k+1}) - \nabla f(x_k)) && \text{(secant condition)} \\
&= \nabla f(x_k).
\end{aligned}$$

This implies that the secant condition ensures that the quadratic model  $m_{k+1}(x)$  accurately reflects the known information at both  $x_{k+1}$  and  $x_k$ , except for the previous function value  $f(x_k)$ .

## 1.2 Superlinear Convergence and the Dennis–Moré Condition

Q-convergence is short for quotient-convergence, referring to convergence rates defined through quotients of successive errors. In particular,

A sequence  $\{x_k\}$  converging to  $x^*$  is said to converge Q-superlinearly if

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0$$

holds. Thus, the contraction factor itself approaches zero, which is stronger than Q-linear convergence with a fixed contraction factor less than one.

The Dennis–Moré condition characterizes when a Newton-like iteration achieves this rate [1, 2]. Consider the iteration

$$B_k s_k = -\nabla f(x_k), \quad x_{k+1} = x_k + s_k$$

and suppose that  $x_k \rightarrow x^*$ ,  $x_k \neq x^*$ ,  $\nabla f(x^*) = 0$ , and that  $\nabla^2 f$  is continuous near  $x^*$  with the matrix  $\nabla^2 f(x^*)$  nonsingular. Then the convergence is Q-superlinear if and only if

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - \nabla^2 f(x^*)) s_k\|}{\|s_k\|} = 0. \tag{1}$$

This is the Dennis–Moré condition. See also [3].

## References

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